far beyond Einstein's relativity. Any story of the world that could incorporate a violation of Bell's inequality would have to be truly strange.

How could Bell possibly prove something so unexpected and so vast in its implications? To fully understand his proof, we'll need more than a roulette wheel—we'll need a whole casino. (If you aren't interested in following the details of the proof, feel free to skip the next section entirely—it won't affect your ability to understand any of the rest of this book. But following through the argument in the next section will give you a greater understanding of how Bell proved what he did.)

A new casino has opened up in the small town of Bellville, California, in the sparsely populated northeastern corner of the state—and it's owned by Ronnie the Bear, who is suspected of having mob connections. Fatima and Gillian, two inspectors from the California Gaming Bureau, head up to Bellville to check out the casino before it opens, because they know Ronnie is probably up to something.

Ronnie's casino floor has an overcomplicated roulette setup, possibly to impress the inspectors. In the center of the room is a large machine, with a chute extending from each side to the roulette tables at either end of the floor. At each of the two roulette tables, there are three roulette wheels, with a smaller spinning dial in the center. In accordance with state law, the roulette wheels only have alternating squares of red and black on them, not numbers—roulette wheels with numbers on them are illegal in the state of California (Figure 7.3). Once Fatima and Gillian are each seated at one of the tables, Ronnie presses a button on the machine, and a roulette ball appears in each of the chutes, rolling toward the tables. The inspectors spin the center dial while the balls are en route, and each roulette ball lands automatically in whichever wheel is selected, eventually settling on a red or black square (Figure 7.4).